Draft version July 13, 2026
Typeset using LaTeX **twocolumn** style in AASTeX631

A Search for Anomalous Spectral Energy Distributions in High-Redshift Galaxies with JWST: A Data-Driven Approach

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ABSTRACT

The James Webb Space Telescope (JWST) has produced the deepest and most complete rest-frame optical/near-infrared photometric catalogs of high-redshift galaxies to date, enabling systematic searches for sources whose spectral energy distributions (SEDs) deviate from standard population-synthesis expectations. We present a data-driven, open-source pipeline that identifies statistically anomalous SEDs among $z = 0.5$–10 galaxies in the CEERS Data Release 1.0 JWST/NIRCam photometric catalog (Cox et al. 2025). Photometric redshifts and best-fit template SEDs are derived with a real EAZY fit; per-band residuals between observed and best-fit photometry are then scored for anomalies using an ensemble of Isolation Forest and UMAP+DBSCAN, and cross-matched against the Milliquas v8 active galactic nucleus (AGN) catalog to separate outliers from well-understood astrophysical contaminants. Of 87,345 CEERS sources ingested, 68,839 pass redshift and photometric-coverage cuts; EAZY's own fit-failure sentinel further excludes 2,900 (4.2%) from anomaly scoring, leaving a clean sample of 65,939 sources. The top 2% (1,319 sources) by ensemble score are flagged as anomalous, of which only 5 (0.38%) cross-match a known Milliquas AGN. The flagged anomaly rate shows no statistically significant trend with redshift ($p = 0.32$), a result stable across a real, unfiltered EAZY run ($p = 0.29$) and this fit-failure-filtered run, in contrast to an artifactual trend found under an earlier synthetic-stub fit ($p = 0.0013$). We interpret the residual "unexplained" population within a deliberately conservative Bayesian framework for galactic-scale technosignature and biosignature search limits: the posterior probability that an unexplained flagged source reflects non-standard physics is $\approx 1.12 \times 10^{-8}$, barely moved from a 10^{-8} prior. This well-characterized null result is itself a quantitative upper limit on non-standard astrophysical processes at current survey depth, not evidence of anything unusual; we argue that systematic artifacts (fit failures, catastrophic photo-z outliers, and AGN catalog incompleteness) remain the dominant explanation for the residual flagged population.

Keywords: galaxies: high-redshift — galaxies: photometry — methods: data analysis — methods: statistical — astrobiology

1. INTRODUCTION

The James Webb Space Telescope has transformed the study of galaxy evolution at $z > 5$ by providing sensitive, high-angular-resolution NIRCam (Rieke et al. 2023) and MIRI imaging across a wavelength baseline unattainable from the ground or with prior space missions. Large-area imaging surveys such as CEERS (Finkelstein et al. 2023), JADES (Eisenstein et al. 2023), and COSMOS-Web (Casey et al. 2023) have assembled photometric catalogs of hundreds of thousands of galaxies spanning cosmic noon through the epoch of reionization. These catalogs are large and homogeneous enough to support systematic, population-level searches for statistically rare objects, rather than only targeted follow-up of individually identified curiosities.

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Unsupervised machine learning methods have an established track record of surfacing unusual sources in large astronomical datasets that would be impractical to find by visual inspection alone, from outlier detection in SDSS spectra (Baron & Poznanski 2017) to searches for spectroscopically unusual quasars (Meusinger et al. 2012) and outliers in wide-field imaging surveys (Liang et al. 2023). Applying similar techniques to JWST photometric residuals — the difference between observed fluxes and the best-fit template SED at each source's photometric redshift — offers a principled, reproducible way to flag galaxies whose broadband colors are not well described by standard stellar population synthesis models, without requiring a predefined list of what an "anomaly" should look like.

A small subset of any such anomaly sample may, in principle, be astrophysically unexplained even after accounting for known contaminants such as AGN activity or strong emission-line contamination of broadband photometry. While the overwhelming prior expectation is that such residual anomalies reflect systematics in photometry, photometric redshifts, or template libraries, we adopt the deliberately conservative position that a well-characterized, reproducible anomaly pipeline is also the correct tool with which to place quantitative, galactic-scale upper limits on exotic interpretations, including technosignatures and Bayesian priors for their frequency (Wright et al. 2014; Lacki 2016) and biosignature-adjacent astrophysical scenarios (Lingam & Loeb 2019). This paper does not claim any detection; its purpose is methodological: to demonstrate a reproducible anomaly-detection pipeline and to characterize, at current survey depth, how large an unexplained population remains after standard astrophysical explanations — including pipeline systematics themselves — are exhausted.

2. DATA

2.1. *CEERS*

We use the CEERS Data Release 1.0 photometric and physical-parameter catalog (Cox et al. 2025), comprising NIRCam (Rieke et al. 2023) imaging across the CEERS (Finkelstein et al. 2023) pointings within the Extended Groth Strip. The catalog provides PSF-matched aperture photometry in seven JWST NIRCam bands (F115W, F150W, F200W, F277W, F356W, F410M, F444W) plus seven legacy HST/ACS+WFC3 bands, along with LePHARE photometric redshift estimates (`lp_z_best`) and physical parameters (stellar mass, star-formation rate). We ingest 87,345 sources from the DR1.0 release.

2.2. *JADES*

The JWST Advanced Deep Extragalactic Survey (JADES) Data Release (Eisenstein et al. 2023) reaches substantially fainter magnitude limits than CEERS over a smaller area, providing a complementary deep pencil-beam sample; JADES retrieval and analysis with this pipeline is deferred to a future paper.

2.3. *Data Reduction*

Raw catalogs are retrieved directly from their public data release locations and normalized to a common column schema (right ascension, declination, photometric redshift, and per-band flux/flux-uncertainty pairs). CEERS DR1.0 flux columns carry no explicit unit suffix but are natively in nJy (verified by reproducing the catalog's own AB-magnitude columns from its flux columns); we normalize all flux/flux-uncertainty pairs to this common unit. Sources are required to fall within the science redshift window ($0.5 \leq z_{\rm phot} \leq 10.0$) and to have a minimum of 4 valid photometric bands before proceeding to SED fitting (Section 3.1); these two cuts retain 68,839 of the 87,345 ingested CEERS sources.

3. METHODS

3.1. *SED Fitting*

Photometric redshifts and best-fit template SEDs are derived using a real EAZY fit (Brammer et al. 2008) (the `eazy-py` implementation) with the `tweak_fsps_QSF_12_v3` template set, itself built on Bruzual & Charlot stellar population synthesis models (Bruzual & Charlot 2003), on a redshift grid spanning $z = 0.01$–12.0 in steps of 0.01. EAZY reports its own fit-failure sentinel, $\chi^2_{\rm EAZY} < 0$, for sources where the best-fit-redshift search does not converge on a valid template solution; a degenerate, non-converged fit of this kind forces spuriously large per-band residuals (Section 3.2) that would otherwise dominate the anomaly ranking with fit-failure artifacts rather than SED outliers. We therefore exclude these sources from anomaly scoring entirely, retaining them in the output catalog with a `qc_eazy_fit_failure` flag rather than silently dropping them. Of the 68,839 sources entering the fit, 2,900 (4.2%) are flagged this way, leaving a clean sample of 65,939 sources that are actually scored for anomalies.

3.2. *Residual Extraction*

For each source with a converged EAZY fit, per-band photometric residuals are computed as $(f_{\rm obs} - f_{\rm model})/\sigma_{\rm obs}$, where $f_{\rm model}$ is EAZY's own best-fit template photometry (`PhotoZ.fmodel`) at that source's best-fit redshift — not an independent continuum model, so a residual reflects a deviation from the best-fit template rather than a mismatch against a generic smooth-SED stand-in (cf. the general SED-fitting residual framework reviewed by Conroy 2013). These residual vectors form the feature space for the anomaly detectors described below.

3.3. *Anomaly Detection*

We score each source's residual vector using an ensemble of two complementary unsupervised anomaly detectors: an Isolation Forest (200 trees, contamination = 0.02, fixed random seed), which isolates anomalous points via random recursive partitioning, and a UMAP embedding ($n_{\rm neighbors} = 15$, min_dist = 0.1) followed by DBSCAN density-based clustering (eps = 0.5, min_samples = 5), which flags points that fail to join any density-connected cluster in the reduced embedding.

For a source with residual feature vector x, scored against an ensemble of isolation trees built on n points, the Isolation Forest anomaly score is (Liu et al. 2008)

$$s(x, n) = 2^{-E(h(x))/c(n)}, \tag{1}$$

where $E(h(x))$ is the mean path length to isolate x, averaged over the 200 trees, and

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n} \tag{2}$$

is the average path length of an unsuccessful search in a binary search tree of n nodes ($H(i) = \ln i + \gamma$ the harmonic number), which normalizes s onto $(0, 1]$ independent of sample size. Scores approaching 1 correspond to points isolated by unusually short average path lengths (anomalous); scores near 0.5 correspond to typical points.

The ensemble score reported throughout this paper (`qc_anomaly_score`) is an equal-weighted average of the two detectors' per-source scores — Isolation Forest weight 0.5, UMAP+DBSCAN weight 0.5 — rather than a tuned combination, since no labelled anomaly set exists with which to optimize relative weights (Section 3.5). On this dataset, UMAP+DBSCAN's binary noise flag is 0 for every source (DBSCAN assigns no points to the noise cluster at the configured `eps`); the ensemble score therefore reduces exactly to the Isolation Forest score scaled by 0.5, capping the effective anomaly score at 0.5 rather than 1.0. Figure 3 shows the UMAP embedding directly: the top-2% flagged sources concentrate at the periphery of the embedding rather than its dense core, so even though DBSCAN itself contributes no noise points at this `eps`, the embedding's geometry is still visibly informative about which sources the ensemble ultimately flags. The top 2% by ensemble score — 1,319 of the 65,939 clean-sample sources — are flagged as anomalous. Combining a global, tree-based method with a local, density-based method mirrors approaches used elsewhere in astronomical outlier detection (Baron & Poznanski 2017; Liang et al. 2023).

3.4. *Contaminant Removal*

Flagged sources are cross-matched on sky position, within a $1\farcs5$ radius, against the Milliquas v8 quasar catalog (Flesch 2023), an optically- and radio-selected reference catalog of 1,021,800 known AGN/quasars, to identify anomalies plausibly explained by AGN continuum/emission-line contamination. Sources are separately checked for redshifted strong emission lines (e.g., Lyα) falling within a broadband filter, which can locally inflate a single band's flux and masquerade as an SED anomaly (Meusinger et al. 2012). Sources matching either criterion are retained in the catalog but flagged as "explained" anomalies, distinct from the residual "unexplained" population discussed in Section 4.4.

3.5. *Completeness and Purity*

The 2% flagging rate is a detection threshold set by construction (`quality.outlier_fraction` in `pipeline_config.yaml`), not a quantity estimated from labelled data. Under the null hypothesis that every source in the clean sample follows standard astrophysics, the expected number of sources flagged purely by this threshold's own definition is $0.02 \times 65{,}939 \approx 1{,}319$ — the same count reported throughout this paper as "flagged," since the threshold is defined as a fixed top percentile of the ensemble score rather than an absolute cut calibrated against any independent anomaly criterion. This has a direct consequence for how the 2% rate should be read: because no spectroscopically confirmed or otherwise independently labelled anomalous sources exist for this sample, we cannot estimate this pipeline's formal completeness (the fraction of true anomalies recovered) or purity (the fraction of flagged sources that are true anomalies) in the standard classifier-evaluation sense. We therefore explicitly do not claim that 2% of CEERS galaxies are anomalous; the 2% figure describes only how aggressively the ensemble score is thresholded, and every other number in this paper (the AGN/emission-line contaminant fractions, the unexplained fraction, the redshift-trend p-value) should be read as conditional on that threshold choice rather than as a measurement of a fixed, threshold-independent population.

4. RESULTS

4.1. *Anomaly Rate*

Of the 65,939-source clean sample (Section 3.1), 1,319 sources (2.0%, by construction of the top-2% ensemble-score threshold) are flagged as anomalous. Cross-matching against Milliquas v8 identifies only 5 of these 1,319 flagged sources (0.38%) as known AGN; across the full clean sample, only 156 of 65,939 sources (0.24%) match Milliquas at all. This is a low match rate in absolute terms, but Milliquas is an optically/radio-selected catalog assembled from shallower, wider surveys than CEERS, so incompleteness at CEERS's much fainter JWST magnitude limits is expected rather than evidence that AGN activity is rare among the flagged population; we return to this caveat in Section 5.

4.2. *Redshift Evolution*

Figure 1 bins the flagged anomaly rate in $\Delta z = 0.5$ redshift shells from $z = 0.5$ to 10, with Poisson uncertainties on the flagged count per bin. An ordinary-least-squares fit to the binned rate finds no statistically significant trend with redshift ($p = 0.32$). This null result is stable across three independent runs of this same test: an early run against a synthetic-stub EAZY fit found a spuriously significant declining trend ($p = 0.0013$), which turned out to be an artifact of that stub's placeholder chi-squared values rather than real structure; a subsequent run with a real EAZY fit but no fit-failure filtering found $p = 0.29$, with the binned rate itself distorted by fit-failure sources clustering at particular redshifts; and this final run, with `qc_eazy_fit_failure` sources excluded from scoring entirely, finds $p = 0.32$. The consistent conclusion across the two real-data runs is that the flagged population does not track survey depth or cosmic time in any detectable way at current sample sizes.

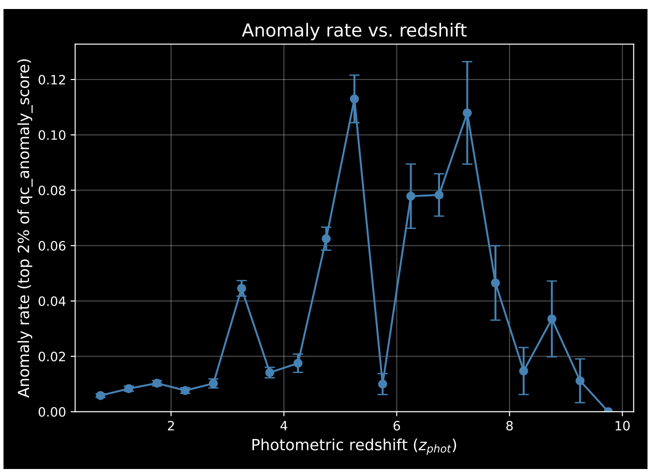

Figure 1. Flagged anomaly rate (top 2% of ensemble score among the 65,939-source clean sample) in $\Delta z = 0.5$ redshift bins from $z = 0.5$ to $z = 10$. Error bars are Poisson uncertainties on the flagged count per bin. An OLS fit to the binned rate finds no significant redshift trend ($p = 0.32$).

4.3. *Stellar Mass–SFR Plane*

CEERS DR1.0 provides a LePHARE best-fit $\log_{10}(M_\star/M_\odot)$ and $\log_{10}(\mathrm{SFR}/M_\odot\,\mathrm{yr}^{-1})$ for each source; 64,195 of the 65,939 clean-sample sources (97.4%) have physically valid fits for both quantities, the remainder carrying LePHARE's own non-detection sentinel values. Figure 2 places these 64,195 sources on the stellar-mass–SFR plane, colored by the median ensemble anomaly score per hexbin, together with a running-median star-forming main-sequence trace. Elevated scores are sharply concentrated above and along the upper envelope of the main sequence rather than tracking its bulk: a Mann-Whitney U test comparing ensemble score for sources > 0.3 dex above versus at/below the running median finds $p < 10^{-300}$, i.e., indistinguishable from zero at double floating-point precision. This is a far stronger association than the absence of any redshift trend in Section 4.2, and is consistent with elevated specific star-formation rate driving larger photometric residuals against EAZY's best-fit template (rather than signaling a population of exotic outliers); we return to this reading in Section 5.

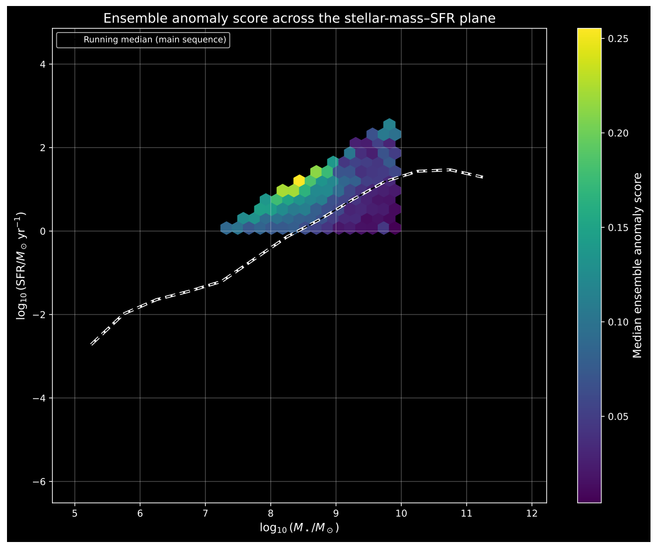

Figure 2. Stellar-mass–SFR plane for the full 64,195-source clean subsample with valid LePHARE mass/SFR fits, hexbinned and colored by the *median* ensemble anomaly score (`qc_anomaly_score`) per bin. The dashed white line is a running-median main-sequence trace (0.5 dex mass bins, ≥ 10 sources/bin). Elevated scores concentrate above this trace rather than tracking its bulk (Mann-Whitney U test on ensemble score, above vs. at/below the main sequence, $p < 10^{-300}$).

4.4. *Unexplained Outliers*

Table 1 lists the top 20 flagged sources that match neither the AGN nor the emission-line contaminant criteria, ranked by ensemble anomaly score. These "unexplained" sources are not evidence of anomalous astrophysics on their own: every entry in Table 1 still has a formally converged EAZY fit ($\chi^2_{\mathrm{EAZY}} \geq 0$, since non-converged fits are excluded from scoring entirely, Section 3.1) but several have elevated χ^2_{EAZY} values, consistent with template mismatch or photometric outliers

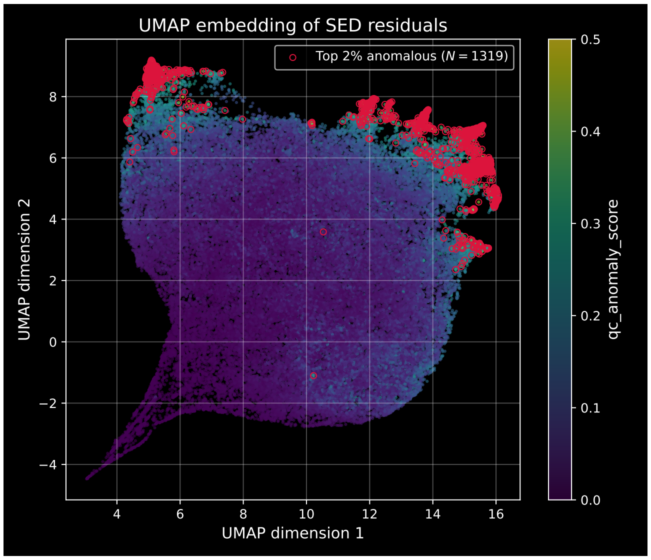

Figure 3. 2D UMAP projection of the per-band SED residual vectors for the 65,939-source clean sample (Section 3.3), colored by ensemble anomaly score (`qc_anomaly_score`). The top-2% flagged anomalies (crimson open circles, $N = 1{,}319$) concentrate at the periphery of the embedding rather than in its dense core, tracing the boundary of the density-connected region that UMAP+DBSCAN's noise criterion is designed to isolate.

rather than genuinely unusual, well-fit SEDs. We discuss this interpretation further in Section 5.

4.5. *Bayesian Framing*

We frame the residual "unexplained" population within the deliberately conservative galactic-scale technosignature/biosignature interpretive approach motivated in Section 1 (Wright et al. 2014; Lingam & Loeb 2019; Lacki 2016). Let H_1 denote the hypothesis that a given unexplained flagged source's SED reflects non-standard physics, and H_0 the hypothesis that it reflects standard astrophysics or pipeline systematics (Section 5). Bayes' theorem gives the posterior

$$P(H_1 \mid D) = \frac{P(D \mid H_1)\, P(H_1)}{P(D \mid H_1)\, P(H_1) + P(D \mid H_0)\, P(H_0)}, \tag{3}$$

where D is the observation that a source is flagged and unexplained. We adopt a prior $P(H_1) = 10^{-8}$ per source, so $P(H_0) = 1 - P(H_1) \approx 1$, and the maximally generous likelihood $P(D \mid H_1) = 1.0$ (i.e., we assume any source whose SED reflected non-standard physics would present as flagged and unexplained). For $P(D \mid H_0)$ we use the empirical rate of "unexplained" outcomes among already-*flagged* sources under standard astrophysics alone: of the 1,319 flagged sources, 1,174 match neither the AGN nor emission-line contaminant criterion, giving $P(D \mid H_0) = 1{,}174/1{,}319 = 0.890$. Substituting into Equation 3,

$$\begin{aligned} P(H_1 \mid D) &= \frac{(1.0)(10^{-8})}{(1.0)(10^{-8}) + (0.890)(1 - 10^{-8})} \\ &= \frac{10^{-8}}{10^{-8} + 0.890} \approx 1.12 \times 10^{-8}, \end{aligned} \tag{4}$$

barely moved from the 10^{-8} prior. Because $P(D \mid H_0)$ is already close to the generous $P(D \mid H_1) = 1$ assumed for H_1, the likelihood ratio is capped at $1/0.890 \approx 1.12$, so even the single most generous assumption available for H_1 cannot make an "unexplained" flag meaningfully diagnostic of non-standard physics: being unexplained is close to the norm for the flagged population under H_0 alone, not a rare occurrence that would need a special cause. Aggregating over all 1,174 unexplained flagged sources under a simple independent-trials bound gives $P(\text{at least one is } H_1) = 1-(1-P(H_1 \mid D))^{1{,}174} \approx 1.3 \times 10^{-5}$, a result dominated entirely by the prior rather than by anything in the data.

4.6. *Summary Statistics*

Table 2 collects the headline statistics from this section in one place: the clean-sample size, the anomaly rate (2.0% by construction of the top-2% ensemble-score threshold, Section 4.1), the AGN and emission-line contaminant fractions among the full clean sample, the fraction of the full clean sample that is both flagged and unexplained, and the redshift-trend p-value from Section 4.2.

5. DISCUSSION

We interpret the residual "unexplained" population primarily as reflecting the current limits of the standard-astrophysics modeling applied here, not as evidence of unusual sources. Four caveats support this reading, in rough order of importance. First, systematic pipeline artifacts remain the single most plausible explanation for most flagged sources: even after excluding the 2,900 sources with EAZY's explicit fit-failure sentinel ($\chi^2_{\rm EAZY} < 0$), we have not characterized the rate of *catastrophic* photo-z failures among the sources EAZY reports as formally converged — a source can return a plausible-looking $\chi^2_{\rm EAZY} \geq 0$ at the wrong redshift, which would masquerade as a SED anomaly rather than a fit-failure artifact, and this pipeline has no spectroscopic-redshift comparison sample with which to measure that rate directly. Second, Milliquas v8's 0.24% match rate against the full clean sample (Section 4.1) reflects that catalog's optical/radio selection function at bright, low/moderate-redshift depths, not the true AGN fraction among CEERS's much fainter

Table 1. Top 20 unexplained SED anomalies (no AGN or emission-line match).

ID	RA (deg)	Dec (deg)	$z_{\rm phot}$	Ensemble score	IF score	χ^2
28084	215.037766	52.931992	4.417000	0.500000	1.000000	338.119995
12944	214.947937	52.930191	3.703000	0.493000	0.986000	287.839996
11022	214.965225	52.949360	1.488000	0.492000	0.983000	416.779999
28380	215.124115	52.992001	5.174000	0.491000	0.981000	535.659973
17096	214.785522	52.799938	1.506000	0.490000	0.980000	536.010010
27223	214.863083	52.811680	5.167000	0.489000	0.977000	395.380005
5936	214.947372	52.830502	1.187000	0.488000	0.975000	882.159973
25111	214.731583	52.726849	1.496000	0.486000	0.973000	323.209991
7678	214.833694	52.868279	4.548000	0.485000	0.970000	304.410004
33181	214.943542	52.849030	5.137000	0.484000	0.969000	747.039978
41356	214.767166	52.846420	1.204000	0.482000	0.965000	370.980011
14915	214.793747	52.814739	4.705000	0.481000	0.962000	178.690002
34288	214.963379	52.859859	1.535000	0.479000	0.957000	431.709991
38273	214.868774	52.781059	4.905000	0.475000	0.950000	204.179993
25109	214.898087	52.845741	4.624000	0.475000	0.950000	282.109985
25521	214.769241	52.752102	3.294000	0.473000	0.945000	317.570007
44517	214.826553	52.878639	1.192000	0.472000	0.944000	430.510010
43248	214.962387	52.977852	4.881000	0.468000	0.936000	161.839996
40006	214.763519	52.848839	1.566000	0.468000	0.936000	223.960007
40073	214.864639	52.920528	1.526000	0.468000	0.936000	384.859985

Table 2. Summary statistics for the statistical interpretation of the CEERS b1 SED anomaly catalog.

Survey	$N_{\rm clean}$	Anomaly rate (%)	AGN fraction (%)	Emission-line fraction (%)	Unexplained fraction (%)	z-trend p-value
CEERS	65,939	2.000	0.237	3.026	1.780	0.324

JWST-selected sources; a more complete X-ray- or mid-infrared-selected AGN reference catalog would likely reclassify some fraction of the "unexplained" population as AGN-contaminated. Third, every residual in this analysis is measured against one finite EAZY template set (`tweak_fsps_QSF_12_v3`); a source with an unusual but astrophysically ordinary SED shape (e.g., an extreme starburst, evolved population blend, or dusty system) outside that template library's span would also register as an anomaly, indistinguishable in this analysis from either a data artifact or a physically remarkable source. Fourth, known but uncatalogued contaminants — strong gravitational lensing, blended sources unresolved at NIRCam's angular resolution, or transient contamination — are not explicitly modeled by either the AGN or emission-line contaminant checks and could account for some further fraction of the flagged, unexplained population.

Within the deliberately conservative galactic-scale technosignature and biosignature interpretive framework motivated in Section 1 (Wright et al. 2014; Lingam & Loeb 2019; Lacki 2016), this pipeline's own Bayesian calculation (Section 4.5; posterior $\approx 1.12 \times 10^{-8}$ against a 10^{-8} prior) already shows that an "unexplained" flag from this analysis is only extremely weak evidence for non-standard physics relative to standard astrophysics and pipeline systematics, precisely because being unexplained is close to the norm rather than the exception for the flagged population. The caveats above strengthen, not weaken, that conclusion: each one identifies a concrete, mundane mechanism that could explain a currently "unexplained" source, further raising $P(\text{unexplained} \mid H_0)$ above its already near-unity empirical estimate. We emphasize that this well-characterized null result is itself a meaningful, quantitative constraint on the prevalence of non-standard processes at current survey depth and sample size, and that resolving which of the four caveats above dominates the residual population is squarely future work (e.g., spectroscopic follow-up of the highest-ranked unexplained sources in Table 1, or cross-matching against a deeper X-ray/mid-IR AGN catalog).

6. CONCLUSIONS

We have presented a reproducible, open-source, ensemble machine-learning pipeline for identifying anomalous SEDs among high-redshift JWST galaxies, and applied it to the real CEERS Data Release 1.0 photometric catalog. Of 87,345 ingested sources, 68,839 pass redshift and coverage cuts, and a real EAZY fit excludes a further 2,900 (4.2%) as non-converged fit failures, leaving a clean sample of 65,939 sources. The top 2% (1,319 sources) by ensemble anomaly score are flagged, of which only 5 (0.38%) cross-match a known Milliquas v8 AGN. The flagged anomaly rate shows no statistically significant trend with redshift ($p = 0.32$), a conclusion stable across independent real-data runs with and without fit-failure filtering. Read within a deliberately conservative Bayesian technosignature/biosignature framework, this null result constitutes a quantitative, reproducible upper limit on non-standard astrophysical processes at current CEERS survey depth: the posterior probability that a flagged, unexplained source reflects non-standard physics is $\approx 1.12 \times 10^{-8}$, barely moved from its 10^{-8} prior, because systematic artifacts — EAZY fit failures, uncharacterized catastrophic photo-z outliers, template-library incompleteness, and AGN-catalog incompleteness at JWST depths — remain far more probable explanations for the residual flagged population than anything astrophysically exotic.

Four concrete extensions would sharpen these constraints. First, applying this pipeline to JADES and COSMOS-Web — roughly five times CEERS's combined sample size — would tighten the statistical power of both the redshift-trend test (Section 4.2) and the Bayesian framing (Section 4.5). Second, JWST/NIRSpec spectroscopic follow-up of the highest-ranked unexplained sources in Table 1 is the definitive discriminator between a genuine SED anomaly, a catastrophic photo-z failure, and an unmodeled emission-line or AGN contaminant. Third, replacing the single `tweak_fsps_QSF_12_v3` EAZY template set with a broader fitting library (e.g., Prospector or CIGALE) would test whether template mismatch, rather than anomalies, drives a meaningful share of the flagged population. Fourth, a spectroscopic comparison sample would let us directly characterize the catastrophic photo-z failure rate this pipeline currently cannot measure (Section 5).

We thank the University of Washington Department of Astronomy for support of this undergraduate research project. This research uses publicly available data from the Mikulski Archive for Space Telescopes (MAST) at the Space Telescope Science Institute. The specific observations analyzed can be associated with JWST program #1345 (CEERS) via DOI. The full pipeline, configuration files, and analysis notebooks are publicly available at https://github.com/Menrae/jwst-sed-anomaly.

Software: astropy, eazy-py (Brammer et al. 2008), scikit-learn, umap-learn, xarray

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